INCH-POUND
MIL-DTL-32572(NAVY)
21 February 2017

DETAIL SPECIFICATION

# ANCHOR WINDLASS, ELECTRO-MECHANICAL, VERTICAL, TWO-SPEED AND VARIABLE-SPEED

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two-speed and variable-speed, electro-mechanical, vertical shaft anchor windlasses on naval ships.

1.2 Classification. Vertical axis, electro-mechanical anchor windlasses with one wildcat, one capstan head (if required), and below deck drive are available in the following types, as specified (see 6.2):

a. Type I – Two-speed

b. Type II – Variable-speed

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

FF-S-92 - Screw, Machine, Slotted, Cross-Recessed or Hexagon Head

COMMERCIAL ITEM DESCRIPTIONS

A-A-52522 - Lining Brake, and Lining Material Brake: Automotive (Metric)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-P-15024/5 - Plate, Identification

MIL-DTL-16392 - Brake, Electro-Mechanical, Naval Shipboard

MIL-DTL-17060 - Motors, Alternating Current, Integral-Horsepower, Shipboard Use

MIL-PRF-23236 - Coating Systems for Ship Structures

MIL-PRF-24635 - Coating Systems, Weather-Resistant, Exterior Use

MIL-PRF-32006 - Programmable Controller, Naval Shipboard

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108 - Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

MIL-STD-130 - Identification Marking of U.S. Military Property

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II - Internally Excited)

MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment

MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-882 - System Safety

MIL-STD-1310 - Shipboard Bonding, Grounding, and other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety

MIL-STD-1399 - Interface Standard for Shipboard Systems

MIL-STD-1472 - Human Engineering

MIL-STD-1474 - Noise Limits

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-299 - Cable Comparison Handbook Data Pertaining to Electric Shipboard Cable

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at http://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMITTEE ON NATIONAL SECURITY SYSTEMS INSTRUCTIONS (CNSSI)

CNSSI No. 1253 - Security Categorization and Control Selection for National Security Systems

(Copies of this document are available online at https://www.cnss.gov/CNSS/issuances/Instructions.cfm.)



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NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-53 - Security and Privacy Controls for Federal Information Systems and Organizations

NIST SP 800-82 - Guide to Industrial Control Systems (ICS) Security

(Copies of these documents are available online at http://www.nist.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

S9074-AR-GIB-010/278 - Requirements for Fabrication Welding and Inspection, and Casting Inspection and Repair for Machinery, Piping, and Pressure Vessels

S9086-TW-STM-010/582 - Naval Ships' Technical Manual Chapter 582, Mooring and Towing

(Copies of these documents are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil/ by searching for the TMIN without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. These documents are available for ordering (hard copy) via the Naval Logistics Library https://nll.ahf.nmci.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/AGMA 6001-E08 - Design and Selection of Components for Enclosed Gear Drives

(Copies of this document are available online at http://webstore.ansi.org/.)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 - Standard Symbols for Welding, Brazing, and Nondestructive Examination

(Copies of this document are available online at www.aws.org.)

ASME

ASME B17.1 - Keys and Keyseats

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM F765 - Standard Specification for Wildcats, Ship Anchor Chain

(Copies of this document are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 4568 - Shipbuilding -- Sea-Going Vessels -- Windlasses and Anchor Capstans

(Copies of this document are available online at www.iso.org.)



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SAE INTERNATIONAL

| | | |
|---|---|---|
| SAE-AMS-STD-595/26270 | - | Interior Haze Gray |
| SAE AS35411 | - | Fittings, Lubrication |
| SAE AS50151 | - | Connectors, Electrical, Circular Threaded, AN Type, General Specification for |

(Copies of these documents are available online at www.sae.org.)

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

| | | |
|---|---|---|
| SSPC-SP 10/NACE No. 2 | - | Near-White Blast Cleaning |

(Copies of this document are available online at www.sspc.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials. The materials shall be capable of meeting all of the requirements specified herein. Unless otherwise specified (see 6.2), cast iron, malleable iron, and semi-steel shall not be used for any part of the anchor windlass.

3.4 Environmental conditions. Unless otherwise specified (see 6.2), anchor windlasses shall be operable and have no loss of lubrication or liquids on ships encountering ±15 degrees of static list, ±5 degrees of static trim, ±45 degrees of dynamic roll (combatants), ±30 degrees of dynamic roll (non-combatants), and ±10 degrees of dynamic pitch. Anchor windlasses shall be operable in temperatures ranging from -40 °F (-40 °C) to 140 °F (60 °C) with relative humidity from 0 to 100 percent.

3.4.1 Noise. Anchor windlass equipment shall not exceed MIL-STD-1474 Grade D airborne sound level acceptance criteria.

3.4.2 Shock. All anchor windlass system equipment shall meet MIL-S-901 Grade B shock requirements. In addition to the ship specification requirements, the portion of windlass equipment required for towing a ship shall meet MIL-S-901 Grade A shock requirements. The wildcat, wildcat brake, and the manual portions of the wildcat brake control are the equipment required for towing a ship. When specified (see 6.2), the anchor windlass system shall meet the shock requirements as identified in MIL-S-901.

3.4.3 Vibration. All rotating machinery shall be balanced in accordance with MIL-STD-167-1 to minimize vibration. The windlass shall operate continuously and reliably during and after exposure to Type I vibration in accordance with MIL-STD-167-1 and shall withstand internally excited vibration in accordance with MIL-STD-167-1 Type II vibration.

3.4.4 Electromagnetic interference (EMI) suppression. All anchor windlass equipment shall be bonded and grounded in accordance with MIL-STD-464 and MIL-STD-1310 to minimize adverse electromagnetic effects and to protect personnel from electrical shock.



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3.5 Design. Anchor windlasses shall consist of one wildcat, one capstan head, one electric motor drive, controls, bedplates, gearing, brakes, control stands, software, a programmable logic controller (PLC) (if used), and all other operating and control accessories suited to handling the weight of an anchor, as specified (see 6.2). Figure 1 provides representative guidance for vertical, electro-mechanical anchor windlasses. The following general design considerations shall be used:

a. The windlass shall meet strength, rigidity, and durability requirements specified herein and withstand the stresses encountered during windlass operation, as specified (see 6.2). The windlass shall form inertial loads associated with starting and stopping of the chain or prime mover during all modes of operation. If required, the maximum size and weight shall not exceed that specified (see 6.2).

b. The design of keys, feathers, splines, nuts, screws, bolts, and collars shall include positive means to prevent their working loose. The use of set screws shall be avoided wherever possible. When set screws are used, means shall be provided to prevent their loosening and working out.

c. The design of all handwheels, levers, and associated members shall withstand a force of 350 pounds applied to the rim or handle when the mechanism is driven against its stops or extreme position without failure or permanent deformation.

d. The minimum wrap angle for the chain wrap on the wildcat shall be 170 degrees when accounting for the design of ship decks, hawsepipes, and chain lockers.

e. The design shall provide compensation for deck deflections of not more than 1 inch in any direction. Where shafts pass through a deck, provisions shall be made to obtain thorough watertightness.



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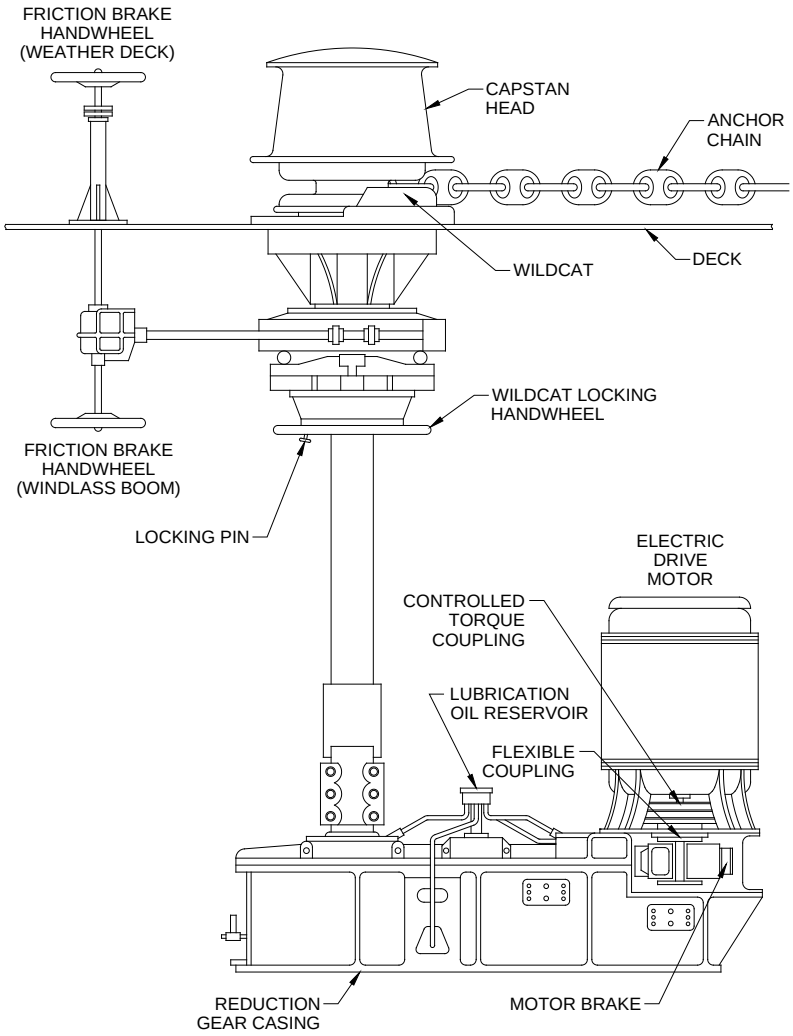



FIGURE 1. Typical vertical, electro-mechanical anchor windlass, representative only.

3.5.1 Welding. Welded fabrication may be used in lieu of castings. Fabrication, welding, and inspection shall be in accordance with S9074-AR-GIB-010/278 and AWS A2.4.

3.5.2 Allowable stress. The strength relationships specified in table I shall be used in design stress calculations.

TABLE I. Allowable stress.

| Design strength | Percent of the tensile yield point |
|---|---|
| Direct shear | 60 |
| Torsional shear | 60 |
| Compression (bearing) | 160 |



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3.5.2.1 Hoisting stress. Under rated normal working load of hoisting the anchor and its chain from 60 fathoms depth, the combined stresses shall not exceed 35 percent of the tensile yield point of the ductile materials used in any part of the equipment.

3.5.2.2 Capstan load stress. Stresses shall not exceed 70 percent of the tensile yield point in any part of the equipment when the capstan head is subjected to a horizontal pull that is equal to the minimum breaking strength of the synthetic rope specified (see 6.2) and acting at mid-height of the capstan head and normal to the axis of the shaft.

3.5.2.3 Brake stress. Stresses shall not exceed 25 percent of the tensile yield point in any part of wildcat brakes or their operating mechanisms and mountings under any of the operating or test conditions specified herein.

3.5.3 Wildcat. The wildcat shall be designed to handle a flash butt welded stud-link chain and connecting links of the size specified (see 6.2). The wildcat shall be designed in accordance with ASTM F765 Type IV. Five-link and three-link sections (eight links total) of the specified size of chain and one detachable link shall be furnished for testing and assuring the fit of the chain on the wildcat casting.

3.5.3.1 Drive. Each wildcat shall be arranged for positive drive through a driving head and locking head device located in the windlass room. The locking head shall be actuated by an operating mechanism of rugged construction that shall positively hold the drive head in the engaged or disengaged position. The entire operation of engaging or disengaging the wildcat drive shall be accomplished by one motion of a lever through an angle not exceeding 60 degrees. The operating gear and lever shall be conveniently located in the windlass machinery room.

3.5.3.2 Separate operability. The wildcat and capstan head may turn together; however, the capstan shall be operable when the wildcat is stationary.

3.5.3.3 Top. The top of the wildcat shall be designed with a step recess on the periphery and shall be fitted with a convex cover plate that has a rim that fits in the step recess. The cover plate shall be constructed of corrosion-resistant metal, bronze, or stainless steel; shall be secured with round head machine screws in accordance with FF-S-92; and shall be gasketed to prevent moisture from entering the interior of the hardware.

3.5.3.4 Support. The wildcat shall be mounted on bearings and be supported such that the drive shafts are not subject to bending. The chain and rope pull operating stress and forces shall be completely taken by the deck bedplates.

3.5.3.5 Key fit. Wildcats shall be shrunk on their respective barrels or shafts and keyed with keys having a drive fit. Keyways shall be in accordance with ASME B17.1 Class 1 fit or better. Keyway corners shall be designed with fillet radii to prevent local stress concentration by operating forces.

3.5.3.6 Speed limiting device. When specified (see 6.2), a speed limiting device shall be furnished. This device shall be positively coupled to the wildcat and shall be designed to limit the speed at which the anchor can be lowered under any condition to a maximum of 1,000 feet per minute. During the free fall condition, the speed control device shall maintain the speed of the anchor’s descent to within a range between 800 to 1,000 feet per minute. A device shall be provided to monitor the chain speed at all times.

3.5.3.7 Chain strippers. Chain strippers shall be provided for freeing the chain from the wildcat when heaving in or veering out.

3.5.4 Capstan heads. Capstan heads shall be of rugged construction and proportioned for the rope size specified (see 6.2).

3.5.4.1 Top. The top of the capstan head shall be designed with a step recess on the periphery and shall be fitted with a convex cover plate that has a rim that fits in the step recess. The cover plate shall be constructed of corrosion-resistant metal, bronze, or stainless steel; shall be secured with round head machine screws in accordance with FF-S-92; and shall be gasketed to prevent moisture from entering the interior of the hardware.

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3.5.4.3 Key fit. Capstan heads shall be shrunk on their respective barrels or shafts and keyed with keys having a drive fit. Keyways shall be in accordance with ASME B17.1 Class 1 fit or better. Keyway corners shall be designed with fillet radii to prevent local stress concentration by operating forces.

3.5.5 Electric motor. The electric motor shall consist of a Service A AC induction motor in accordance with MIL-DTL-17060. A nonstandard motor or permanent magnet motor that is in accordance with MIL-DTL-17060 is also acceptable. Wildcats and capstans shall be reversible and equally operable in either direction of rotation. Type I windlasses shall have two-speed motors. Type II windlasses shall have variable-speed motors. The mean speed shall meet the requirements of 3.8. The driven chain speed shall not exceed twice the required mean speed.

3.5.5.1 Motor control. The electric motor shall be controlled by pushbutton master switches located on the windlass deck as well as in the motor room below. “Start”, “Stop”, and “Emergency Run” elements shall be provided in the master switches. Electric motor controllers and controller characteristics shall be in accordance with MIL-DTL-2212 and as specified (see 6.2).

3.5.5.2 Gear reducer. High-speed gear reduction between the electric motor and the windlass shafting, if required, shall be mounted on antifriction bearings, fully enclosed, and positively lubricated.

3.5.5.3 Motor couplings. Flexible all-steel couplings shall be fitted between the electric motor and the high-speed reduction gears unless the gears are mounted within the electric motor casing. Where no high-speed reduction is required, the coupling shall be between the motor and the drive shaft. Flexible all-steel couplings shall be provided between the motor and the drive shaft of the main reduction gear. The coupling arrangements shall be such that any unit may be removed with a minimum disturbance of the other equipment.

3.5.5.4 Vertical shafts. Slip joints or couplings shall be provided for vertical drive shafts where deflection or warping of decks may cause misalignment of the shafts.

3.5.5.5 Electrical equipment interface. All major electrical equipment shall operate from a 440 volts AC (VAC), 60 Hertz, three-phase high resistance ground, Type I power system having steady-state and transient characteristics in accordance with MIL-STD-1399 and MIL-DTL-917. All other electrical equipment, including those utilizing 115 VAC single- or three-phase power, shall be in accordance with the electrical interface requirements set forth in MIL-STD-1399.

3.5.6 Control stations. Controls shall be arranged for convenient and safe operation of the equipment. Controls shall remain in the position set by the operator. Controls shall be as follows:

a. Control stand windlass level: controls and indicators.

b. Control station motor level: controls and indicators.

c. Handwheel windlass and motor levels: operation of the associated wildcat brake (may be directly linked).

d. Remote controls, if required: as specified (see 6.2).

3.5.6.1 Control functions. At a minimum, the control functions shall include:

a. A vertically-mounted master switch that provides direction (hoist/lower) and speed selection and incorporates a dead man switch that returns to neutral when no force is applied.

b. A white indicator light for power on.

c. An emergency stop pushbutton switch.

d. White engage and disengage indicator lights for the clutch.

e. A lamp test functionality.

f. A lamp dimmer switch.

g. A chain counter with a digital display, if applicable.

h. A red overload protection indicator light.

i. A station and control indicator light.



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3.5.6.2 Windlass level controls. The windlass level control design shall provide a clear view of the operation of the associated windlass.

3.5.6.3 Motor level controls. The motor level shall contain both a motor controller and a logic controller. The logic controller may be a PLC in accordance with MIL-PRF-32006. The motor level shall also include a handwheel in close proximity to the associated wildcat brake. These stations shall be used during maintenance and outside normal operations. There shall be a means to turn power on and off and to select the control station to be used on the motor controller rather than at each control station. When power is applied to the motor controller, the station that the user would like to operate shall be able to be selected. The station in control shall have a lit indicator light on its control station while the non-selected station shall have a non-lit indicator light.

3.5.6.3.1 Emergency stop pushbutton. The pushbutton shall be red in color and surrounded by a protective bezel to prevent inadvertent operation. The pushbutton shall be located away from other indicators/operators such that inadvertent operation is avoided. The emergency pushbutton shall cause loss of all power to the drivetrain and cause all action to stop immediately. Reset of the emergency pushbutton shall not cause automatic restart; instead, manual restart from motor control panel shall be required.

3.5.6.3.2 Lamp dimmer. The dimmer shall control all indicators including power indication and overload indication. In the full “on” condition, the lamps and indicators shall be visible in daylight. Near the “off” condition, lamps and indicators shall be capable of night-time operation.

3.5.6.4 Water protection. The motor controller and PLC (if used) shall be in drip-proof enclosures. The control stands shall be watertight. Drip-proof and watertight protection designations shall be in accordance with MIL-DTL-2036.

3.5.6.5 Cabling. Electrical cabling furnished with equipment should be selected using MIL-HDBK-299 as guidance.

3.5.6.6 Cable connectors. All electrical switches, connections, and lighting fixtures shall be protected by watertight fittings. Cable connectors shall be in accordance with SAE AS50151.

3.5.6.7 Junction boxes. Junction boxes shall be provided as part of each control station and control cabinets to gain access to leads from logic, safety, and door switches for troubleshooting purposes.

3.5.6.8 Shielding. Shielding on wires and cable shall be secured to prevent contacting or shorting of exposed current-carrying parts where necessary. To prevent fraying, the shielding shall have secured or braided ends.

3.5.6.9 Mechanical aspects. The control mechanical design shall consist of joints, shafts, and gears mounted on antifriction bearings and arranged for minimum lost motion. Bearings and gears shall be completely enclosed. Bearings shall be spaced and located to prevent whipping and excessive deflection of control shafting. Steel tubing may be used for shafting where practicable to save weight and secure torsional stiffness. Universal joints or splined slip joints shall be provided where necessary to allow for deflection and working of the ship structure. Universal joints shall be secured to control shafts with keys locked by self-locking set screws or by other means. Stuffing boxes shall be furnished where control shafting passes through watertight decks or bulkheads. Roll pins, tapered pins, and bolts shall not be used in joints of the control shafting.

3.5.6.10 Handwheel effort. The control handwheel shall require not more than 30 pounds of effort at the wheel rim for normal operation. The turns from zero stroke to full stroke shall not exceed 1½ turns.

3.5.7 Bedplates. Bedplates shall be of cast or welded steel construction so as to maintain alignment of all parts, to transmit external forces to the ship’s structure without objectionable deflection, and to resist the shock forces specified in 3.5.2. Bedplates to be installed on weather decks or in the anchor windlass room shall be watertight with no pockets or hollows in which water or debris may collect. The underside of bedplates shall be finished to a plane surface for watertight mounting on foundations.



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3.5.7.1 Wildcat bedplates. Each wildcat shall be mounted on a bedplate, which may consist of one piece or two pieces bolted together. When specified (see 6.2), the bedplate shall contain chain fairleads from the hawsepipe and the chain pipe to the wildcat. Fairleads shall be formed to an easy radii and shall take up not less than three complete links of the chain. The height of fairleads and chain pipe deck bolsters shall be such as to deliver the chain to the horizontal center line of the wildcat between whelps. The location of the chain pipe deck bolsters shall allow for handling the chain in and out of the wildcat. The angle of rake and the details for connecting the chain pipes to the chain pipe deck bolsters shall be as specified (see 6.2). The radii of flared mouth of the chain pipe deck bolsters leading to the wildcats shall be such that the chain binding is minimized. When specified (see 6.2), chain pipe deck bolsters shall be fitted with portable watertight metal bucklers fastened by means of hinged drop bolts with non-detachable bronze nuts.

3.5.7.2 Power unit bedplates. Each power unit shall consist of an electric motor, reduction gearing, if required, control equipment, and related accessories. Each power unit shall be mounted on a bedplate in the motor room. Bearing pads for units mounted on bedplates shall be machined to flat surfaces and fitted bolts shall be used to secure the units to bedplates. Tap bolts shall not be used.

3.5.8 Gears. Gears, if used, shall be designed in accordance with ANSI/AGMA 6001-E08 and transmit the full torque of the windlass.

3.5.8.1 Gear lubrication. Gearing shall be pressure-lubricated by a power-driven oil pump or shall be lubricated by a splash system with upper bearings pressure lubricated. A strainer shall be provided in a power-driven oil pump lubrication system.

3.5.8.2 Gear cases. Gearing shall be fully enclosed in an oiltight case arranged for the ready removal of gears and bearings. Cases shall be provided with bolt- or screw-fastened oiltight inspection covers, fills, and drain openings equipped with oil sample valves. Oiltight inspection covers shall be provided in sufficient number to allow visual examination of the teeth on all gears, permit cleaning of the gear case bottom, and allow inspection of all oil lubricating pump strainers. Cases shall have a means for determining and limiting the oil level within the case without removing the cover. Recommended oil level shall be indicated on a level indicator dipstick or sight glass. An air vent that is baffled to prevent the loss of oil under operating conditions and of a type that excludes foreign matter shall be provided for each gear case. Lifting eyebolts shall be installed as necessary on gear cases. Gear case joints shall be oiltight. Gear cases of 5-gallon capacity and larger shall be fitted with quick-disconnect suction and return lines to permit installation and fluid maintenance.

3.5.9 Brake. A hand brake shall be provided. Power-assist and electric brakes shall be as specified (see 6.2).

3.5.9.1 Hand brake. The wildcat shall be fitted with an externally contracting, hand-operated band brake located below the weather deck and controlled by a screw-type handwheel. Manual operation shall be from a horizontal handwheel located in the anchor handling area(s) near the control station(s). Each brake handwheel control station shall have a mechanical pointer that indicates engagement or disengagement. The brake shall stop and hold the anchor and chain in no more than the below listed rates per depth:

a. 15 fathoms out: stop within 3 fathoms.

b. 30 fathoms out: stop within 3 fathoms.

c. 45 fathoms out: stop within 4 fathoms.

d. 60 fathoms out: stop within 5 fathoms.

3.5.9.2 Brake bands. The wildcat brake bands shall be lined with non-asbestos, non-burn brake band lining in accordance with A-A-52522 Type I. Means shall be provided for lifting and keeping brake bands free of their drums when brakes are released. Necessary supports and guides that ensure smooth and easy operation of brake bands and operating mechanisms shall be furnished.



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3.5.9.3 Power assist. When specified (see 6.2), anchor windlasses that handle anchors weighing 30,000 pounds and over or designed to handle anchor chains of 3½-inch wire diameter and larger shall be provided with power-assist control as well as manual screw-type handwheel control for applying the wildcat brake. The power-assist control shall be located on the manual handwheel control. Setting of the brake by power assist, when utilized, shall be accomplished by a partial turn of the handwheel that is used to provide manual operation. Means shall be provided for adjusting the brake application force under power-assisted operation so as to avoid a slamming on effect of the brakes. When selected by the operator, each wildcat brake control power-assist or manual screw-type handwheel shall apply the wildcat brake in all modes of windlass operation without requiring any change or adjustment to windlass components.

3.5.9.4 Electric brake. When specified (see 6.2), electric brakes with brake characteristics in accordance with MIL-DTL-16392 shall be provided. To prevent the anchor and chain from overhauling the anchor windlass, the brakes shall be arranged to set upon loss of power or set when power is not applied to the electric motor. The electric brake shall have sufficient capacity to stop and hold 150 percent of the weight of the anchor and 80 fathoms of chain while being lowered at full speed under control of the power unit. Supports and covers shall ensure that oil or grease does not leak/drip onto brake surfaces.

3.5.9.5 Wildcat movement. Brakes for all windlasses shall be designed to meet all requirements specified herein regardless of the wildcat's directional movement.

3.5.9.6 Manual brake effort. The effort required at the rim of the brake handwheel to set the brake shall not exceed 100 pounds of force under any condition.

3.5.9.7 Brake handwheel. The deck band brake control handwheel shall be horizontally mounted on a pedestal and located with full regard to protect operating personnel from the whip of the bitter end of the chain. The height above deck and dimensions of the handwheel shall be in accordance with MIL-STD-1472. The handwheel shall turn clockwise to set the brake and shall be marked to indicate the direction of rotation for setting and releasing the brake. The number of turns from set position to complete off position shall not exceed three turns. The brake operating gear and mechanism shall be non-overhauling and self-locking. An indicator showing an "off" and "on" position shall be provided at each control station.

3.5.9.8 Brake control rod joints. Joints shall be such that the mating components are secured with two keys that are 90 degrees apart. The keys shall be secured positively with self-locking set screws or by other means.

3.5.10 Software. Any and all software, firmware, and programs necessary to configure, operate, monitor, maintain, and program the system and any programmable devices contained therein shall be provided. All system software shall be documented, verified, and certified as specified (see 6.2).

3.5.10.1 Programming. The programming, software, firmware, and user programs shall be provided in electronic format and shall be compatible with use on a portable computer for the purpose of viewing and editing the programs and for uploading and downloading the programs to/from the system via a serial ethernet communication port. There shall be no proprietary protocols or communication ports. The source code for all software shall be provided. All necessary programming tools and communication cables shall be provided.

3.5.10.2 Program access control. Software and programs shall be capable of incorporating password-based access controls to limit the upload, download, deletion, or modification of software and programs to personnel inputting the correct authorization password. An ethernet communication port shall be provided on the controller enclosure via a lockable, hinged cover to allow for programming, monitoring, and troubleshooting of the programmable control system with the enclosure door closed.

3.5.10.3 Program retention. All software and user programs necessary to operate, monitor, and maintain the system shall be stored in non-volatile memory within the system. Removal or interruption of electrical power shall not cause software and user programs to be erased, altered, or corrupted. Energy storage devices shall not be used to meet this requirement.



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3.5.11 Cybersecurity. The anchor windlass cybersecurity (see 6.4) design shall follow requirements for a secure industrial control system (ICS) in accordance with NIST Special Publications (SPs) 800-53 and 800-82 and CNSSI No. 1253. ICS includes supervisory control and data acquisition (SCADA) systems, distributed control systems (DCS), and other control system configurations, such as PLCs, which are often found in the industrial control sectors. The anchor windlass shall contain interfaces for external digital communications, whether used or unused. Cybersecurity shall be in accordance with Appendix A.

3.5.11.1 Cybersecurity strategy. The anchor windlass cybersecurity strategy shall be as specified (see 6.2) and approved by NAVSEA.

3.5.11.2 PLC. Commercial PLCs shall be designed in accordance with MIL-PRF-32006 (see 6.2). The motor controller shall be in accordance with MIL-DTL-2212. All electrical equipment shall be in accordance with MIL-STD-461, MIL-STD-167-1, MIL-DTL-917, and MIL-STD-108, except as specified in this section.

3.5.12 Lubrication. Means for lubrication of all working parts from readily accessible locations shall be provided. This equipment is known to be inactive for extended periods of time; therefore, splash and transfer lubrication methods shall not be used. If a power-driven oil pump is provided, it shall be furnished for spray-type lubrication of gears and bearings of the main gear drive between the wildcat and capstan shafts. If spray-type lubrication systems are provided, they shall be active when equipment controls are powered. Pressure grease lubrication shall be provided for all other bearings. All bearings in gear housings not completely lubricated by oil in the housing shall be fitted with grease fittings for pressure grease lubrication in accordance with SAE AS35411. Large bearings, such as wildcat and capstan head bearings, shall be provided with several lubricant fittings but not less than three fittings equally spaced on the periphery of the bearing. Each lubricating fitting shall be installed at a safe distance from rotating components of the anchor windlass and, if possible, at one common place.

3.6 Reliability and maintainability. Units shall meet or exceed the following requirements:

a. Mean time between (mission critical) failure: 1,000 operational cycles, or as specified (see 6.2).

b. Unit design life: 30 years, or as specified (see 6.2).

c. Mean time to repair: 6.0 hours, or as specified (see 6.2).

d. Minimum time between overhauls: 250 operational cycles, or as specified (see 6.2).

e. Statistical confidence of 80 percent, or as specified (see 6.2).

3.7 Performance characteristics.

3.7.1 Anchor hoisting. Without exceeding the full-load rating of the motor, the windlass shall hoist an anchor and chain of the specified weight and size (see 6.2) from a depth of 60 fathoms at an average rate of at least 5 fathoms per minute and from a depth of 100 fathoms up to 60 fathoms at no specified speed.

3.7.2 Capstan. Capstan heads shall develop a rope speed of at least 125 feet per minute with the specified size of rope (see 6.2) at no load condition. The capstan shall also withstand a static load equal to the minimum breaking strength of the specified size of rope using the head as a bitt at the mid-height of the head and normal to its axis. The capstan shall be able to pull rated load at rated speed. Under either load, the combined stresses in any part of the equipment shall not exceed specified allowable stresses (see 3.5.2).

3.7.3 Brake performance. Brakes shall be of ample size and strength to permit performance of specified brake tests without combined stresses in any part of the brake or operating mechanism exceeding the allowable stresses (see 3.5.2).

3.7.3.1 Hand brake. The hand brake shall meet the requirements of 3.5.9.1.

3.7.3.2 Electric brakes. If required (see 6.2), electric brakes shall meet the performance requirements of 3.5.9.4.

3.7.4 Chain handling. The windlass system shall be capable of executing towing and mooring procedures in accordance with S9086-TW-STM-010/582.



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3.8 Human factors. The anchor windlass system design shall accommodate normal operation by personnel in the 5th through 95th percentile in accordance with MIL-STD-1472.

3.9 Safety.

3.9.1 Operational safety. The design shall include guards and protective items to minimize the potential for personnel injury in accordance with MIL-STD-882.

3.9.2 Loss of power. Manual operation of the windlass system for lowering the anchor shall be possible in the event of electrical power loss. Failure of the power source of any power-operated drive mechanism or control shall not result in damage to the anchor windlass equipment, jeopardize the safety of personnel, or result in uncontrolled movement of the equipment and load.

3.9.3 Control system safety features. Safety feature components and controls shall not be exposed to damage under normal operating conditions or be exposed to encourage possible tampering by operating personnel where such tampering could cause the safety feature to become disabled.

3.9.4 Overload protection. The control system shall provide protection for the motor in accordance with MIL-DTL-2212.

3.10 Dimensions and weight. The size of the chain, anchor, and rope shall be as specified (see 6.2).

3.11 Paints and color. Coatings shall be applied shortly after manufacture and cleaning. Metal to be coated shall have surfaces prepared to “near-white” metal in accordance with SSPC-SP 10/NACE No. 2. The abrasive blast surface profile shall be 2 to 4 mils.

3.11.1 Painted surfaces. Anchor windlass external surfaces shall be coated with two coats of epoxy in accordance with MIL-PRF-23236 Type VI, Class 5 and two top coats of silicone alkyd in accordance with MIL-PRF-24635 Type III, Class 2, Grade C. The silicone alkyd shall be interior haze gray, number 26270 of SAE-AMS-STD-595. Unless otherwise specified (see 6.2), coatings shall be applied in accordance with the coating manufacturer’s product data sheet.

3.11.2 Unpainted surfaces. Bearing and machined surfaces or interior “oil-wetted” surfaces shall not be painted. Corrosion-resistant materials shall not be painted.

3.12 Identification plates. Identification plates and other designated marking plates shall be in accordance with MIL-P-15024/5. All plates, including operating and lubrication charts exposed to the weather, shall be fabricated from corrosion-resistant steel and shall be Type A, B, or C. Charts shall be mounted by fully bedding with adhesive and attached with corrosion-resistant steel fasteners. Identification and marking shall be in accordance with MIL-STD-130.

3.13 Workmanship. The windlasses and components shall be free from imperfections that might impair their serviceability or present hazards to operating personnel. The external finish shall be clean, smooth, and provide complete coverage. There shall be no signs of blistering, corrosion, discoloration, scrapes, gouges, or voids in the finish. Welds shall be free of harmful defects such as cracks, undercuts, voids, or gaps. There shall be no sharp or ragged edges, corners, burrs, or protrusions that might create a safety hazard to personnel. There shall be no dents or loose, damaged, bent, or broken parts or fasteners. There shall be no defects that might reduce the reliable operational capability or life of the equipment. There shall be no signs of fluid leakage. Piping and cabling shall be supported by clamps or hangers. Cabling and wiring insulation shall not be nicked, gouged, cut, or otherwise damaged. Safety guards, covers, shields, and warning plates shall be appropriately installed and marked.

## 4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).



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4.2 First article inspection. First article inspection shall be performed on electro-mechanical anchor windlasses when a first article sample is required (see 3.1). This inspection shall include the tests as specified in table II and the requirements of ISO 4568.

4.3 Conformance inspection. Conformance inspection shall include the tests as specified in table II.

TABLE II. First article and conformance inspection and testing.

| Characteristic | | Requirement | First article | Conformance |
|---|---|---|---|---|
| Noise | | 3.4.1 | 4.6.1.3 | -- |
| Shock | | 3.4.2 | 4.6.1.4 | -- |
| Vibration | | 3.4.3 | 4.6.1.5 | -- |
| EMI | | 3.4.4 | 4.6.1.6 | -- |
| High temperature | | 3.4 | 4.6.1.7 | -- |
| Low temperature | | 3.4 | 4.6.1.8 | -- |
| Humidity | | 3.4 | 4.6.1.9 | -- |
| Shop | Pre-checks/No load | 3.8, 3.9 | 4.6.2.1 | 4.6.2.1 |
| | Normal operation | 3.7.1 | 4.6.2.2 | 4.6.2.2 |
| | Wildcat hand brake | 3.5.9.1, 3.7.3.1 | 4.6.2.3 | 4.6.2.3 |
| | Electric motor brake | 3.5.9.4, 3.7.3.2 | 4.6.2.4 | 4.6.2.4 |
| | Wildcat chain | 3.5.3 | 4.6.2.5 | -- |
| Dock | Pre-checks/No load | 3.8, 3.9 | 4.6.3.1 | 4.6.3.1 |
| | Normal operation | 3.7.1 | 4.6.3.2 | 4.6.3.2 |
| | Wildcat hand brake | 3.5.9.1, 3.7.3.1 | 4.6.3.3 | 4.6.3.3 |
| | Electric motor brake | 3.5.9.4, 3.7.3.2 | 4.6.3.4 | 4.6.3.4 |
| | Capstan static load | 3.7.2 | 4.6.3.5 | -- |
| | Capstan normal load | 3.7.2 | 4.6.3.6 | 4.6.3.6 |
| Deep sea | Hoisting | 3.7.1 | 4.6.4.1 | 4.6.4.1 |
| | Wildcat hand brake | 3.5.9.1, 3.7.3.1 | 4.6.4.2 | 4.6.4.2 |
| | Electric brake | 3.5.9.4, 3.7.3.2 | 4.6.4.3 | 4.6.4.3 |
| Chain handling | | 3.7.4 | 4.6.4.4 | 4.6.4.4 |
| Reliability and maintainability | | 3.6 | 4.6.5 | 4.6.5 |
| Software | | 3.5.10 | 4.6.6 | 4.6.6 |
| Cybersecurity | | 3.5.11 | 4.6.7 | 4.6.7 |



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4.4 Test conditions. For the tests specified in 4.6, the equipment is not required to operate under full-load conditions for more than 1 hour in a 4-hour period. Unless otherwise specified (see 6.2), environmental conditions shall meet MIL-STD-810 Coastal/Ocean, Exposed (for weather deck machinery), and Sheltered (for interior machinery) criteria.

4.5 Examination. Anchor windlasses shall be visually examined for compliance with the requirements specified in 3.5 through 3.5.8.2, 3.5.3.1 through 3.5.8, 3.5.9.2, 3.5.9.3, 3.5.9.5 through 3.5.9.8, 3.5.12, and 3.10 through 3.13. This element of inspection shall encompass all visual examinations and dimensional measurements.

4.6 Tests.

4.6.1 Environmental tests.

4.6.1.1 Equipment to be tested. The following equipment, which may experience degradation or operational impacts due to environmental effects, shall be tested:

a. Motor.
b. Control stations.
c. PLC (if used).
d. Controls.
e. Chain counter.
f. Other electric cabinets and relays.

4.6.1.2 Readings to be taken. The following readings shall be taken during no-load and load tests:

a. Revolutions per minute (RPM) of wildcat shaft.
b. Volt, amperes, kilowatt input, and power factor.
c. Oil pressure and temperature.
d. RPM of electric motor.
e. Time (start, stop, and duration).
f. Hoisting speed.
g. Performance as defined herein of the hand brake.
h. Temperature of gearings and bearings before, during, and after tests.
i. Performance as defined herein of electric brakes.

4.6.1.3 Airborne noise test. The airborne noise shall be measured in accordance with the instrumentation and procedures of MIL-STD-1474 with the exception that noise levels at the measurement positions shall be averaged. Motors shall be tested while operating at no load with rated voltage and frequency applied.

4.6.1.4 Shock test. Components shall be tested for high-impact shock in accordance with MIL-S-901.

4.6.1.5 Vibration test. The anchor windlass shall be tested in accordance with MIL-STD-167-1 Type I, ranging from 4 hertz (or lowest attainable frequency) through 50 hertz, and Type 2.

4.6.1.6 EMI test. The equipment and associated electronics shall be tested for EMI in accordance with MIL-STD-461.

4.6.1.7 High temperature test. The equipment shall be subjected to a high temperature test in accordance with MIL-STD-810 Method 501.4, Procedure II. The cyclic temperature test shall be performed.

4.6.1.8 Low temperature test. The equipment shall be subjected to a low temperature test in accordance with MIL-STD-810 Method 502.4, Procedure I.

4.6.1.9 Humidity test. The equipment shall be subjected to a humidity test in accordance with MIL-STD-810 Method 5.7.6.



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4.6.2 Operational shop tests.

4.6.2.1 No-load test. With the wildcat engaged, each anchor windlass shall operate continuously with no external loading for at least ½ hour. Direction shall be alternated several times such that each direction receives 50 percent of the total time. Repeat this test for each motor speed or through the achievable range for variable-speed designs.

4.6.2.2 Normal operation test. The anchor windlass shall be subjected to a test that simulates normal operation to determine conformance to the requirements of 3.9. A brake system may be used to provide the necessary torque resistance equal to 30 fathoms of chain and anchor. The speed shall be at an average of 5 fathoms per minute without exceeding the full-load rating of the motor or controller. The time duration shall be a minimum of 10 minutes.

4.6.2.3 Wildcat hand brake test. The wildcat brake shall be subjected to a wildcat brake torque capacities test to demonstrate that the equipment can resist the calculated breaking torque without deformation or damage. The torque applied shall be relative to a 15 fathom drop of the anchor and chain. The wildcat brake shall be set using the handwheel with the drive motor brake released. A calibrated tension device shall be used to ensure the load required to the anchor rigging and tension device. After the load is stabilized, the time duration shall be a minimum of 10 minutes.

4.6.2.4 Electric motor brake test. If electric brake installation is specified (see 6.2), the test of motor brake torque capacities shall demonstrate that the equipment can resist the calculated torque without deformation or damage. The drive unit brake shall stop and hold 150 percent of the combined weight of the anchor and 80 fathoms of chain while being lowered at full speed under control of the drive unit. A calibrated tension device for the load required shall induce a tension load to the anchor rigging and tension device. After the load is stabilized, the time duration shall be a minimum of 10 minutes.

4.6.2.5 Wildcat chain testing. The five-link and three-link sections shall be removed from the bitter end of the anchor chain. The five-link and three-link sections shall be reinstalled on the bitter end of the anchor chain after completing a wildcat fit check using the detachable link furnished for the test.

4.6.3 Dock tests. Each windlass shall be tested dockside after installation on board the ship as specified herein.

4.6.3.1 No-load test. Each windlass unit shall be run for 30 minutes in each direction with the wildcat disconnected.

4.6.3.2 Normal operation test. Each windlass unit shall be tested by hoisting and lowering the anchor to 30 fathoms in 33 fathoms or more of water at an average rate of not less than 5 fathoms per minute continuously for four complete cycles. This test may be conducted in a lesser depth of water, provided that sufficient weights are added so that the total weight lifted is equivalent to the weight of the anchor and 30 fathoms of chain in the water. This test shall also cover inspection of the operation of the chain on the wildcat and in the chain pipe. Without exceeding the full-load rating of the motor, each windlass shall be tested by hoisting a weight equal to 100 fathoms of chain plus the anchor weight (with compensatory weight added, if necessary [as described above]) a distance of 30 fathoms at no specified speed.

4.6.3.3 Wildcat hand brake test. Each unit shall be tested with weight added as described above so that the total weight is equivalent to the weight of the anchor and 30 fathoms of chain in the water. The anchor shall be held in the stowage position in the hawsepipe by the hand brake while the locking head is disconnected. The brake shall then be released and the anchor let go until 15 fathoms of chain are out or until the anchor is within 2 fathoms of the bottom in testing conditions where the water is less than 15 fathoms deep at which time the brake shall be applied. The brake shall stop and hold the anchor and chain within 2 fathoms travel or before the anchor grounds. The brake test shall be repeated until the wildcat hand brake demonstrates successful operation.

4.6.3.4 Electric motor brake test. If electric brake installation is specified (see 6.2), a weight equal to 100 fathoms of chain plus the anchor weight shall be lowered as in normal operation until the anchor is within 3 fathoms of the bottom at which time the electric motor shall be de-energized and the electric brake shall function. The electric brake shall stop and hold the load before the anchor grounds.



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4.6.3.5 Capstan static load test. Each capstan head shall be tested statically by using the head as a bitt to hold a load equal to the breaking strength of the synthetic rope for 10 minutes as specified with no deformation allowed. The capstan driving mechanism is not required to raise a weight equal to the static rating.

4.6.3.6 Capstan normal load test. Each capstan head with the wildcat disconnected shall be used to demonstrate by tension dynamometer or by other means that the capstan is capable of pulling its rated load at the rated speed. No part of the equipment shall show any sign of permanent distortion or undue wear after these tests.

4.6.4 Deep sea tests. When the ship is on sea trials and in a depth of at least 60 fathoms of water but not exceeding 65 fathoms, anchor-handling arrangements and gear shall be tested as specified herein to demonstrate satisfactory operation in all respects under service conditions.

4.6.4.1 Hoisting test. With the anchor bending shackle just outside the hawsepipe, each anchor shall be independently lowered under power until the 60-fathoms detachable link reaches the waterline and then hoisted to the stowage position in the hawsepipe. Lowering and hoisting of the anchors and chain to 60 fathoms shall be at an average rate of not less than 5 fathoms per minute.

4.6.4.2 Wildcat hand brake test. With the anchor in the hawsepipe and with the locking head disconnected, the anchor shall be dropped by releasing the hand brake. Anchor descent shall be stopped at 15, 30, 45, and 60 fathoms of chain out of the hawsepipe by operating the brake handwheel without additional leverage. The brake shall stop and hold the anchor and chain in not more than 3 fathoms of travel with 15 and 30 fathoms out, in not more than 4 fathoms of travel with 45 fathoms out, and in not more than 5 fathoms of travel with 60 fathoms out.

4.6.4.3 Electric brake test. If electric brake installation is specified (see 6.2), electric brakes shall be tested while lowering the anchor at the maximum speed under power. When 60 fathoms of chain have been let out, the electric motor shall be de-energized and the electric brake shall function. The brake shall stop and hold the anchor and chain before the anchor grounds.

4.6.4.4 Chain handling test. The installed windlass shall rig the ship to be towed in accordance with S9086-TW-STM-010/582. The windlass shall demonstrate the capability to handle the chain while executing this rigging.

4.6.5 Reliability and maintainability test.

4.6.5.1 Reliability. The anchor windlass unit shall meet the requirements of 3.6. If required (see 6.2), new designs or major redesigns shall, at a minimum, have a failure mode, effects, and criticality analysis performed. During testing, a failure reporting, analysis, and corrective action, a system shall be used to record failures and document corrective actions. The duration of a test and the allowable number of failures for the purpose of demonstrating a mean time between failure of 1,000 cycles and statistical confidence of 80 percent (or as specified in 6.2) shall be in accordance with table III.

TABLE III. Reliability test data.

| Duration of demonstration test (hours) | Maximum allowable failures |
|---|---|
| 1,600 | 0 |
| 3,000 | 1 |
| 4,300 | 2 |
| 5,500 | 3 |



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One complete cycle comprises lowering an anchor and chain 60 fathoms at a speed of 1,000 feet per minute followed by hoisting the anchor and chain at a rate of 5 fathoms per minute or approximately 15 minutes per cycle. The anchor and chain can be replaced with a simulated load placed on the capstan head. Parts and components that are subject to wearing out can be replaced at cyclic intervals consistent with the manufacturer's recommendations. A failure in this context is any event that results in the need for corrective maintenance or repair in order for the anchor windlass unit to lower and raise the anchor and chain or simulated load.

4.6.5.2 Maintainability. The maintenance procedures, removal of parts for repair, and the time required for accomplishment shall be verified during first article testing. The demonstration procedures in MIL-HDBK-470 may be used.

4.6.6 Software test. Software, if used, shall be tested as specified (see 6.2).

4.6.7 Cybersecurity plan. Cybersecurity requirements shall be verified in accordance with Appendix A.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The electro-mechanical windlass is intended for use in lowering and raising an anchor and chain when anchoring and to reduce the use of hydraulic machinery aboard naval ships. The capstan head is intended for use in handling lines and for warping service.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type of anchor windlass, including capstan option (see 1.2).

c. When first article is required (see 3.1).

d. When cast iron is to be used if required (see 3.3).

e. Environmental condition requirements (see 3.4 and 4.4).

f. Acquisition information and data requirements as identified in section 6.2 of MIL-S-901, as applicable and any additional shock requirements if necessary (see 3.4.2).

g. Weight of anchor to be handled (see 3.5, 3.5.a, and 3.7.1).

h. Maximum permissible size and weight of windlass, if required (see 3.5.a and 3.10).

i. Rope size and strength to be used by capstan, if capstan is specified (see 3.5.a, 3.5.2.2, 3.5.4, 3.7.2, and 3.10).

j. Chain size to be handled (see 3.5.a, 3.5.3, 3.7.1, and 3.10).

k. Speed limiting device (see 3.5.3.6).

l. Motor and controller characteristics (see 3.5.5.1).

m. Remote operability requirements if necessary (see 3.5.6.d).

n. Bedplate fairleads (see 3.5.7.1).

o. Chain pipe and deck bolsters details (see 3.5.7.1).



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p. Power-assist and electric motor brakes if necessary (see 3.5.9, 3.5.9.3, 3.5.9.4, 3.7.3.2, 4.6.2.4, 4.6.3.4, and 4.6.4.3).

q. Software and software testing requirements (see 3.5.10 and 4.6.6).

r. Cybersecurity strategy (see 3.5.11.1).

s. PLC (if used) (see 3.5.11.2).

t. Reliability and maintainability demonstration and analysis requirements as well as additional ship specific requirements (if necessary) (see 3.6.a, 3.6.b, 3.6.c, 3.6.d, 3.6.e, and 4.6.5.1).

u. Additional coating requirements if necessary (see 3.11.1).

v. Test conditions if other than specified herein (see 4.4).

w. Packaging requirements (see 5.1).

x. Additional documentation (see 6.3).

y. Security categorization results (see A.3.1).

z. Threat analysis (see A.3.2).

aa. Additional cybersecurity requirements (see A.4.2).

6.3 Recommended documentation. The following documentation should be requested as part of the acquisition of an anchor windlass unit, when specified (see 6.2):

a. Test reports.

b. Calculations.

c. Technical manuals.

d. Product drawings.

e. Provisioning technical documentation.

6.4 Cybersecurity. Prevention of damage to, protection of, and restoration of computers, electronic communication systems, electronic communication services, wire communication, and electronic communication, including information contained therein, to ensure its availability, integrity, authentication, confidentiality, and non-repudiation.

6.5 Subject term (key word) listing.

Brakes

Capstan

Chain

Cybersecurity

Deck machinery

Wildcat



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# ANCHOR WINDLASS CYBERSECURITY REQUIREMENTS

## A.1 SCOPE

A.1.1 Scope. This appendix covers anchor windlass cybersecurity requirements. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

## A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in Appendix A of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in Appendix A of this specification, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMITTEE ON NATIONAL SECURITY SYSTEMS INSTRUCTION (CNSSI)

CNSSI No. 1253 - Security Categorization and Control Selection for National Security Systems

(Copies of this document are available online at https://www.cnss.gov/CNSS/issuances/Instructions.cfm.)

DEPARTMENT OF DEFENSE ISSUANCES

DoDI 8500.01 - Cybersecurity

DoDI 8510.01 - Risk Management Framework (RMF) for DoD Information Technology (IT)

(Copies of these documents are available online at www.dtic.mil/whs/directives/.)

FEDERAL INFORMATION PROCESSING STANDARDS (FIPS) PUBLICATION

FIPS Pub 199 - Standards for Security Categorization of Federal Information and Information Systems

(Copies of this document are available online at http://csrc.nist.gov/publications/PubsFIPS.html.)

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

NIST SP 800-53 - Security and Privacy Controls for Federal Information Systems and Organizations

NIST SP 800-53A - Assessing Security and Privacy Controls in Federal Information Systems and Organizations: Building Effective Assessment Plans

(Copies of these documents are available online at http://www.nist.gov.)

NAVAL SEA SYSTEMS COMMAND (NAVSEA) MANUAL

NAVSEA 9400.2-M - NAVSEA PIT-Control System Cybersecurity Implementation Manual

(Copies of this document are available online at www.navsea.navy.mil.)

A.2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.



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## A.3 CYBERSECURITY REQUIREMENTS

A.3.1 System categorization (SC). The generalized format for expressing the security category is confidentiality, integrity, availability, and specific impact where the acceptable values for potential impact are low, moderate, or high. In accordance with FIPS Pub 199, the security category of the anchor windlass is determined as:

SC anchor windlass = {(confidentiality, MODERATE), (integrity, HIGH), (availability, HIGH)}

When specified (see 6.2), if a specific anchor windlass design is determined to have different impact values, the new impact values shall be reviewed and approved by NAVSEA.

A.3.2 Threat analysis. When specified (see 6.2), an analysis of threats shall drive implementation details of the anchor windlass, both physically and with respect to cybersecurity, to mitigate those threats and the impacts of attacks. The analysis of threats shall identify the following and be documented:

a. Anchor windlass assets (e.g., control consoles, human-machine interfaces [HMIs], backbone networks, control processors, communication interfaces, input/output [I/O], networks, and other such assets)

b. Anchor windlass threats, both accidental and malicious (e.g., advanced persistent threats [APTs] and threats on system availability, integrity, and authenticity)

c. Risk to operations and assets, individuals, other organizations, and the Nation posed by the threats

d. Mitigation considerations

A.3.3 Security controls. The security controls from the baseline, based on system categorization in A.3.1, shall be tailored from NIST SP 800-53 using CNSSI No. 1253. The tailored-out security controls shall be provided with technical justification. At a minimum, the following security control families shall be considered in selecting security controls for the anchor windlass. However, the security control selection process shall be fully executed in accordance with DoDI 8510.01 by considering the threats and risks identified in A.3.2.

a. Access control
   (1) The principle of least privilege
   (2) Automatic account lockout after unsuccessful logon attempts
   (3) Remote access capability
   (4) Prohibition of wireless access

b. Audit and accountability
   (1) Generation of time stamps for audit records
   (2) Protection of audit information from unauthorized access, modification, and deletion

c. Configuration management
   (1) Access restrictions for any changes including upgrades and modifications to hardware, software, or firmware components
   (2) Documentation of configuration settings that can be changed in hardware, software, or firmware components
   (3) Disabling of unused or unnecessary physical and logical ports and protocols
   (4) Software whitelisting of a deny-all, permit-by-exception policy to allow the execution of authorized software

d. Identification and authentication
   (1) User identification and authentication
   (2) Password-based authentication including minimum password complexity, lifetime of password, and prohibition of password reuse
   (3) Hardware token-based authentication such as PKI-based tokens (CAC)
   (4) Change of vendor default authenticator prior to delivery/installation
   (5) Device identification and authentication using Media Access Control



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e. Physical and environmental protection
   (1) Use of alarms to monitor physical access point
   (2) Use of lockable physical enclosures
f. System and services acquisition
   (1) Structure hardware, software, and firmware to control access with least privilege
g. System and communications protection
   (1) Failure in a known safe state
h. System and information integrity
   (1) Malicious code protection mechanisms including anti-virus signature definitions and secure coding practices
   (2) Intrusion detection for network connections
   (3) Integrity checks of software, firmware, and information at system startup
   (4) Code authentication of software or firmware prior to installation
   (5) Check of the valid syntax and semantics of inputs

A.3.4 Implementation. The selected security controls for the anchor windlass design from A.3.3 shall be implemented in accordance with DoDI 8500.01, DoDI 8510.01, and NAVSEA 9400.2-M.

A.3.5 Assessment. The implemented security controls for the anchor windlass design in A.3.4 shall be assessed in accordance with DoDI 8510.01 and NIST SP 800-53A.

A.4 CYBERSECURITY VERIFICATION

A.4.1 Verification. The requirements specified in A.3 shall be verified (see table II). Any failure to meet these requirements is cause for rejection of the anchor windlass.

A.4.2 Additional specified cybersecurity requirements. Any additional cybersecurity requirements as specified in 6.2 shall be verified. Any failure to meet these requirements is cause for rejection of the anchor windlass.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 3950-2016-001)

Review activities:
Navy – AS, CG
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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